

Structural bioinformatics

MIB2: metal ion-binding site prediction and modeling server

Chih-Hao Lu^{1,2}, Chih-Chieh Chen³, Chin-Sheng Yu⁴, Yen-Yi Liu⁵, Jia-Jun Liu¹, Sung-Tai Wei⁶ and Yu-Feng Lin^{7,*}

¹The Ph.D. Program of Biotechnology and Biomedical industry, China Medical University, Taichung 404333, Taiwan, ²Institute of Bioinformatics and Systems Biology, National Yang Ming Chiao Tung University, Hsinchu 300193, Taiwan, ³Institute of Medical Science and Technology, National Sun Yat-sen University, Kaohsiung 804201, Taiwan, ⁴Department of Information Engineering and Computer Science, Feng Chia University, Taichung 407102, Taiwan, ⁵Department of Public Health, China Medical University, Taichung 406040, Taiwan, ⁶Department of Neurosurgery, China Medical University Hospital, Taichung 404332, Taiwan and ⁷Department of Medical Laboratory Science and Biotechnology, Asia University, Taichung 413305, Taiwan

*To whom correspondence should be addressed.

Associate Editor: Pier Luigi Martelli

Received on March 18, 2022; revised on June 26, 2022; editorial decision on July 22, 2022; accepted on July 27, 2022

Abstract

Motivation: MIB2 (metal ion-binding) attempts to overcome the limitation of structure-based prediction approaches, with many proteins lacking a solved structure. MIB2 also offers more accurate prediction performance and more metal ion types.

Results: MIB2 utilizes both the (PS)² method and the AlphaFold Protein Structure Database to acquire predicted structures to perform metal ion docking and predict binding residues. MIB2 offers marked improvements over MIB by collecting more MIB residue templates and using the metal ion type-specific scoring function. It offers a total of 18 types of metal ions for binding site predictions.

Availability and implementation: Freely available on the web at <http://bioinfo.cmu.edu.tw/MIB2/>.

Contact: yflin@asia.edu.tw

Supplementary information: [Supplementary data](#) are available at *Bioinformatics* online.

1 Introduction

Interactions between proteins and specific metal ions are essential processes in many physiological activities (Andreini *et al.*, 2004). Identifying metal ion-binding (MIB) sites is vital for understanding the mechanism underlying the biological interactions of MIB proteins. Computational methods for identifying MIB sites can be approximately grouped into sequence-based, e.g. TargetS (Yu *et al.*, 2013) and MIBonSite (Qiao and Xie, 2019) and structure-based, e.g. IonCom (Hu *et al.*, 2016) and BioMetAll (Sanchez-Aparicio *et al.*, 2021) methods. Although structure-based approaches are generally more accurate than sequence-based methods, the limitation of structure-based approaches is that many proteins lack an existing solved structure. This study attempted to address this limitation by expanding upon the MIB site prediction and docking server (Lin *et al.*, 2016). For proteins lacking solved structures, MIB2 gains predicted structures by the (PS)² modeling method (Chen *et al.*, 2009) or by collecting from the AlphaFold Protein Structure Database (Varadi *et al.*, 2022). The MIB applications have been used to validate the potential of the designed site to coordinate metal ions (Aires *et al.*, 2019) and guide the fabrication of heterogeneous biocatalysts based on bioinorganic nanoflowers (Escobar *et al.*, 2017). The

ability of MIB2 to construct a 3D protein structure and provide a larger choice of specific types of metal ions enables users to more conveniently investigate protein functions and design novel binding sites for proteins.

2 Methods

As an updated version of MIB, MIB2 offers a total of 18 types of metal ions (Ca²⁺, Cu²⁺, Fe³⁺, Mg²⁺, Mn²⁺, Zn²⁺, Cd²⁺, Fe²⁺, Ni²⁺, Hg²⁺, Co²⁺, Cu⁺, Au⁺, Ba²⁺, Pb²⁺, Pt²⁺, Sm³⁺ and Sr²⁺), supportive of binding site predictions. Both prediction systems use the fragment transformation method (Lu *et al.*, 2006) to compare regional structures between the query proteins and MIB templates. All MIB templates are extracted from the Protein Data Bank structures of protein and metal ion complexes. Compared with MIB, MIB2 generates more MIB templates for each metal ion type (Supplementary Table S1). As the MIB2 prediction server incorporates the (PS)² method, users can submit queries on either protein sequences or structures. For proteins lacking 3D structures, MIB sites can be identified after structure prediction by protein sequence. MIB2 modifies the scoring function by adding the weighted

Table 1. Comparisons of predictive sensitivities (%) of MIB and MIB2 at a >95% specificity threshold

Metal ion	MIB	MIB2	Metal ion	MIB	MIB2
Ca ²⁺	48.9	61.0	Hg ²⁺	40.9	59.4
Cu ²⁺	85.6	84.2	Co ²⁺	64.5	72.6
Fe ³⁺	85.4	91.6	Cu ⁺	76.6	86.3
Mg ²⁺	37.0	43.7	Au ⁺	NA	44.3
Mn ²⁺	61.4	76.4	Ba ²⁺	NA	34.2
Zn ²⁺	71.1	77.1	Pb ²⁺	NA	41.5
Cd ²⁺	41.2	41.4	Pt ²⁺	NA	41.8
Fe ²⁺	92.9	92.5	Sm ³⁺	NA	47.1
Ni ²⁺	70.7	75.2	Sr ²⁺	NA	45.4

NA: Not applicable.

BLOSUM62 substitution matrix and the weighted contact number of each metal ion, enabling the evaluation of binding potentials for residues in the query proteins. Thus, the docked position of the metal ion may be acquired by inverse transformation from the aligned template. The [Supplementary Methods](#) provide details and [Supplementary Figure S1](#) shows the workflow of our prediction method.

3 Results

3.1 A performance comparison between MIB and MIB2

MIB2 benefits from more protein structures becoming available since the development of MIB. For each type of metal ion, the numbers of binding templates in MIB2 range from 1.24 to 10.72 times the numbers in MIB. [Table 1](#) and [Supplementary Table S2](#) compare the predictive performances of MIB2 and MIB for 18 types of MIB predictions. The accuracies of all 18 metal ions ranged from 93.6% to 95.2% in MIB2; marked improvements from the MIB server were noted with nine of the metal ions, which exhibited increases in sensitivities of between 4.5% and 18.5%. For three of the metal ions (Cu²⁺, Cd²⁺ and Fe²⁺), differences between the MIB and MIB2 prediction systems were <1.5%, which may be due to a sampling error because of the different datasets. Sensitivities of the six newly added MIB site predictions in MIB2 (Au⁺, Ba²⁺, Pb²⁺, Pt²⁺, Sm³⁺ and Sr²⁺) ranged from 34.2% to 47.1%.

3.2 The binding preferences of residue and atom types

The metal ion type-specific scoring function in MIB2 contains two newly added terms, the weighted BLOSUM62 substitution matrix and the weighted contact number of the docked metal ion. The former is used to calculate similarity amongst aligned residues between query proteins and templates. The latter account for the binding preferences of atom types. [Supplementary Figures S2 and S3](#) show the heatmaps of the MIB residues and atom frequencies. Preferential binding occurred with certain types of residues and atoms. For example, Ca²⁺ tends to bind with oxygens, whether in the backbone or sidechain of a residue. This suggests that the residue type does not dominate Ca²⁺ binding. The *Clostridium thermocellum* xylanase, Xyn10B (PDB ID: 1H6Y, chain B), contains five Ca²⁺ binding residues: THR14, GLU16, LYS39, GLU42 and ASP149 ([Xie et al., 2001](#)). Amongst these, LYS39 is a positively charged residue, but binds with Ca²⁺ by a backbone oxygen. Whereas the prediction failed with MIB without a matching template, it was successful with MIB2. LYS39 was aligned with six residue types, i.e. GLY, ILE,

ARG, LYS, GLU and THR, in different templates ([Supplementary Table S3](#)). For 12 metal ion types, the correlation coefficient between the Shannon entropy of MIB residue frequencies and improvement in sensitivity was 0.47 ([Supplementary Fig. S4](#)). Improvements in sensitivity were greater for the metal ion types with higher entropies, indicating that the weighted BLOSUM62 substitution matrix and the weighted contact number successfully regulate the binding score by residue and atom preference.

3.3 Conclusion

Of the 18 types of metal ions in MIB2, 10 had sensitivities exceeding 50%; 7 had sensitivities exceeding 75%. MIB2 is a state-of-the-art server with more accurate and reliable prediction performance than MIB.

Funding

This work was supported by the China Medical University, Taiwan [CMU110-ASIA-07 to C.-H.L.].

Conflict of Interest: none declared.

Data availability

The data underlying this article are available in the web server and in its online supplementary material.

References

- Aires, A. *et al.* (2019) A simple approach to design proteins for the sustainable synthesis of metal nanoclusters. *Angew. Chem. Int. Ed. Engl.*, **58**, 6214–6219.
- Andreini, C. *et al.* (2004) A hint to search for metalloproteins in gene banks. *Bioinformatics*, **20**, 1373–1380.
- Chen, C.C. *et al.* (2009) (PS)2-v2: template-based protein structure prediction server. *BMC Bioinformatics*, **10**, 366.
- Escobar, S. *et al.* (2017) Understanding the functional properties of bio-inorganic nanoflowers as biocatalysts by deciphering the metal-binding sites of enzymes. *J. Mater. Chem. B*, **5**, 4478–4486.
- Hu, X. *et al.* (2016) Recognizing metal and acid radical ion-binding sites by integrating *ab initio* modeling with template-based transfers. *Bioinformatics*, **32**, 3260–3269.
- Lin, Y.F. *et al.* (2016) MIB: metal ion-binding site prediction and docking server. *J. Chem. Inf. Model.*, **56**, 2287–2291.
- Lu, C.H. *et al.* (2006) The fragment transformation method to detect the protein structural motifs. *Proteins*, **63**, 636–643.
- Qiao, L. and Xie, D. (2019) MIONSite: ligand-specific prediction of metal ion-binding sites via enhanced AdaBoost algorithm with protein sequence information. *Anal. Biochem.*, **566**, 75–88.
- Sanchez-Aparicio, J.E. *et al.* (2021) BioMetAll: identifying metal-binding sites in proteins from backbone preorganization. *J. Chem. Inf. Model.*, **61**, 311–323.
- Varadi, M. *et al.* (2022) AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Res.*, **50**, D439–D444.
- Xie, H. *et al.* (2001) Clostridium thermocellum Xyn10B carbohydrate-binding module 22-2: the role of conserved amino acids in ligand binding. *Biochemistry*, **40**, 9167–9176.
- Yu, D.J. *et al.* (2013) Designing template-free predictor for targeting protein-ligand binding sites with classifier ensemble and spatial clustering. *IEEE/ACM Trans. Comput. Biol. Bioinform.*, **10**, 994–1008.